

Government Engineering College Palakkad
Department Of Electronics & Communication Engineering

ECQ 413 SEMINAR PRESENTATION

Power Management Integrated Circuits (PMICs) Design & Applications

Seminar Guide : **Prof. Jithesh C P**
Asst.Professor ECE

HARIKRISHNAN K S - PKD23EC034

Date: 11-08-2026

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01 Introduction

- Modern electronic systems (smartphones, laptops, IoT devices, wearables, EVs, and medical equipment) require multiple stable voltage levels with low power consumption.
- A **Power Management Integrated Circuit (PMIC)** integrates multiple power-management functions into a single chip.
- PMICs manage, convert, distribute, and monitor electrical power efficiently.
- They improve battery life, reduce power consumption, enhance system reliability, and simplify circuit design.

02 Objectives of Power Management Integrated Circuits (PMICs)

- Provide stable and regulated power supply.
- Improve power conversion efficiency.
- Extend battery life by reducing power consumption.
- Generate multiple voltage levels from a single power source.
- Protect against overvoltage, overcurrent, and overheating.
- Reduce PCB size by integrating multiple power functions into one IC.

03 What is PMICs ?

- A **Power Management Integrated Circuit (PMIC)** is an analog or mixed-signal integrated circuit that controls the generation, conversion, regulation, sequencing, monitoring, and protection of power supplied to electronic devices.

04 Need for PMICs

- Modern electronic devices require multiple operating voltages.
- A battery provides a single voltage (typically **3.7 V**).
- A PMIC converts this into multiple regulated voltages with minimal power loss.

Components	Operating Voltage
CPU Core	0.8 V
Memory	1.2 V
Sensors	1.8 V
USB	5 V
Display	3.3 V

05 Design Requirements

- **High Power Efficiency:** Reduces energy loss during power conversion.
- **Low Quiescent Current:** Extends battery life in standby mode.
- **Stable Output Voltage:** Provides reliable power to system components.
- **Fast Transient Response:** Maintains voltage during rapid load changes.
- **Low Ripple & Noise:** Prevents interference in analog and RF circuits.
- **Built-in Protection:** Protects the system from electrical faults.
- **Compact Size & High Integration:** Enables smaller, lightweight electronic devices.

06 Block Diagram

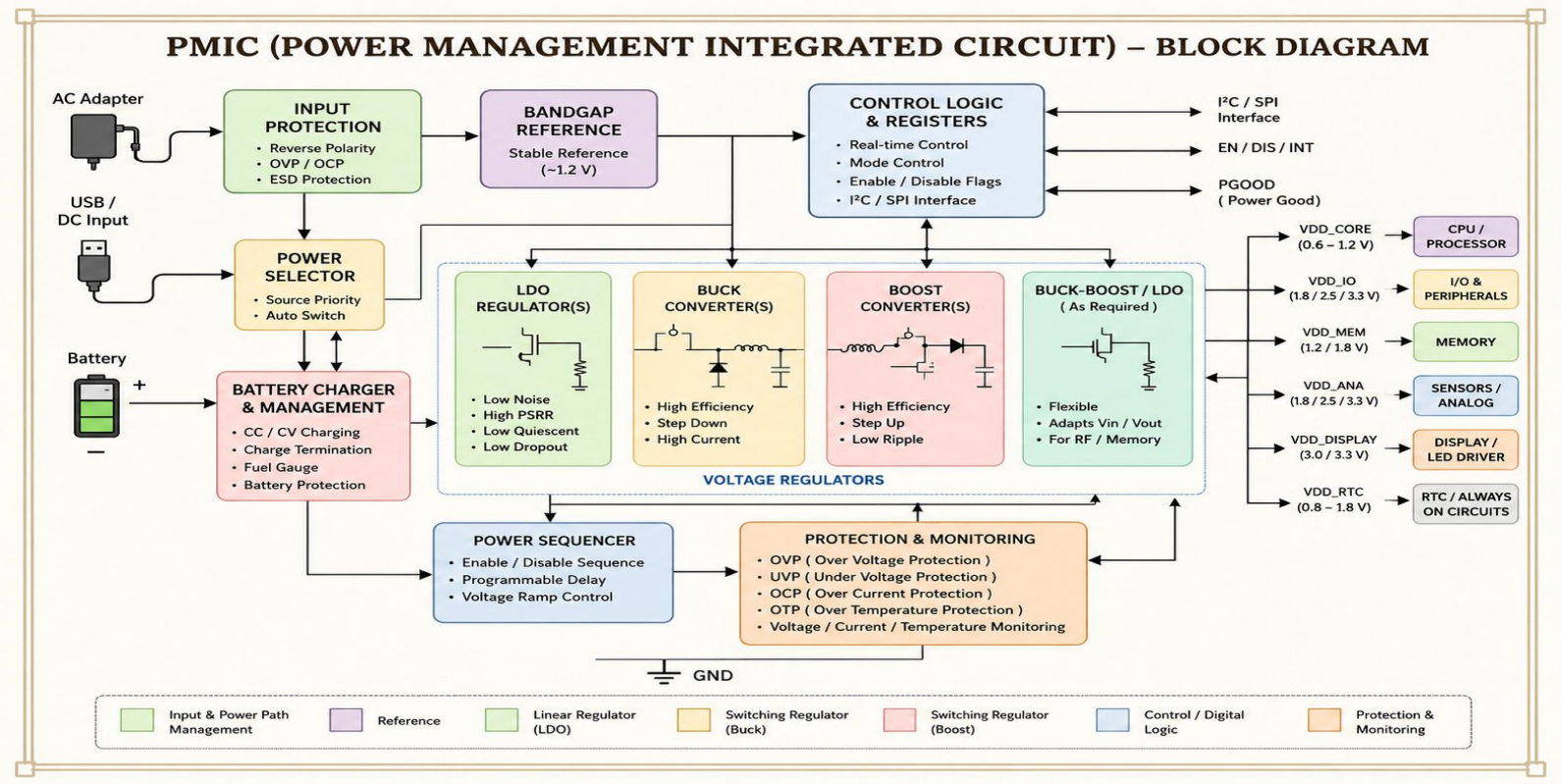


Fig 1. PMIC Block Diagram; *Ref: Chatgpt*

07 Design Blocks

a) Input Protection Circuit

The **Input Protection Circuit** is the first stage of the PMIC. It protects the IC from abnormal electrical conditions before power enters the internal circuitry.

Working

- Monitors battery, USB, or adapter input.
- Detects reverse polarity.
- Protects against overvoltage and overcurrent.
- Suppresses ESD.
- Prevents faults from reaching the PMIC.

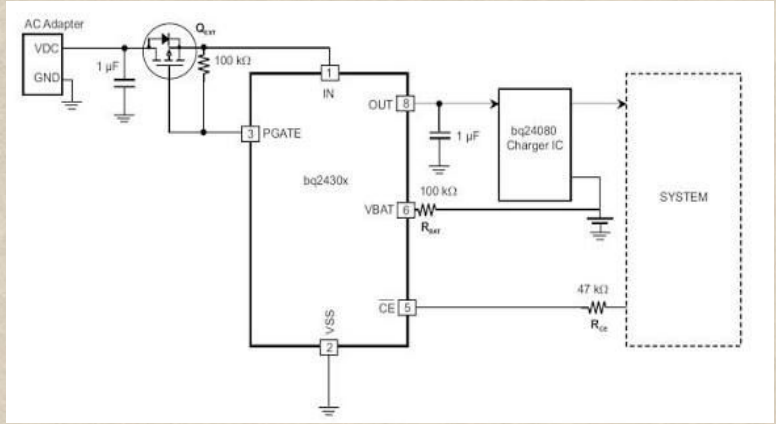


Fig 2. Input Protection Circuit
Ref: Electronic stack exchange

b) Power Selector

A **Power Selector Circuit** is a circuit that automatically selects the most appropriate power source, such as a **battery, USB, or AC adapter**, to supply power to the system. It ensures uninterrupted operation by switching between available power sources while preventing reverse current and enabling battery charging when external power is present.

Working

- Detects USB/adapter and battery power.
- Prioritizes USB/adapter when available.
- Charges the battery from the external supply.
- Automatically switches to battery when needed.
- Prevents reverse current and ensures uninterrupted PMIC power.

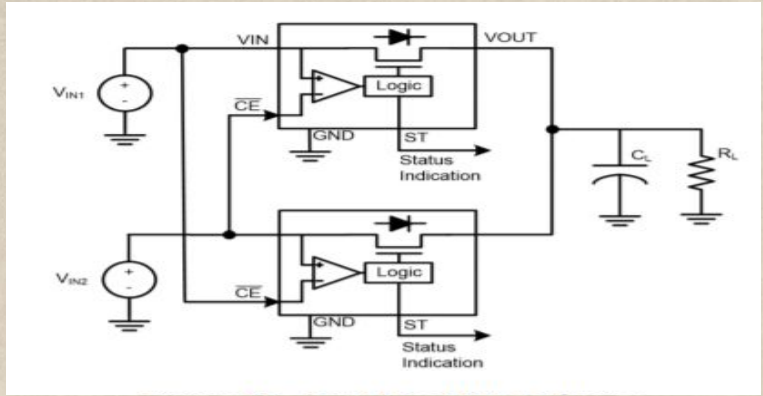


Fig 3. Power Selector Circuit; *Ref : e2e*

c) BandGap Voltage Reference

A bandgap reference generates a **stable reference voltage (≈1.2 V)** that remains nearly constant despite changes in **supply voltage, temperature, and process variations**. It is a fundamental block in PMICs, LDOs, ADCs, DACs, and voltage regulators.

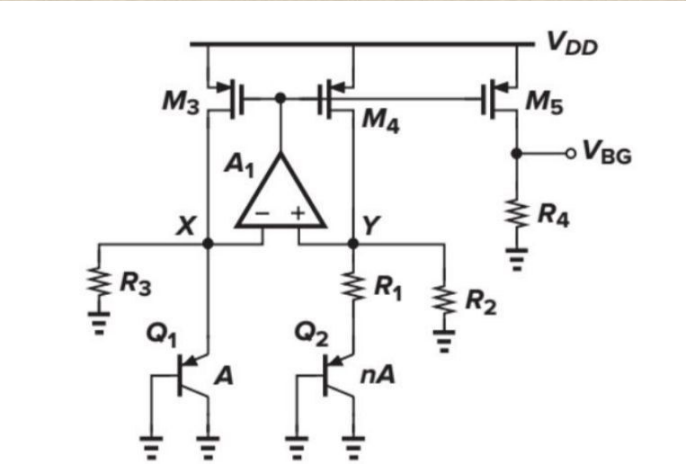


Fig 5. Bandgap Voltage Reference Circuit
Ref: istambuluniversity,BGR - Alper Kurt

Circuit Components

- **Q1 & Q2:** Generate temperature-dependent voltages (CTAT and PTAT).
- **A1:** Keeps the voltages at nodes **X** and **Y** equal.
- **M3, M4 & M5:** PMOS current mirror that provides bias current.
- **R1–R4:** Generate PTAT current and set the output voltage.
- **VBG:** Stable reference voltage output (~1.2 V).

Working

- **A1** forces the voltages at **X** and **Y** to be equal.
- **Q1** and **Q2** generate **PTAT** and **CTAT** voltages.
- **R1–R4** combine these voltages to cancel temperature variations.
- **M3–M5** provide the required bias current.
- The circuit produces a **stable reference voltage (VBG ≈ 1.2 V)** that is nearly independent of temperature and supply voltage.

d) Control Logic and Registers

The **Control Logic and Registers** block is the digital control unit of a PMIC. It manages the operation of all internal power blocks by configuring settings, monitoring system status, and controlling power sequencing.

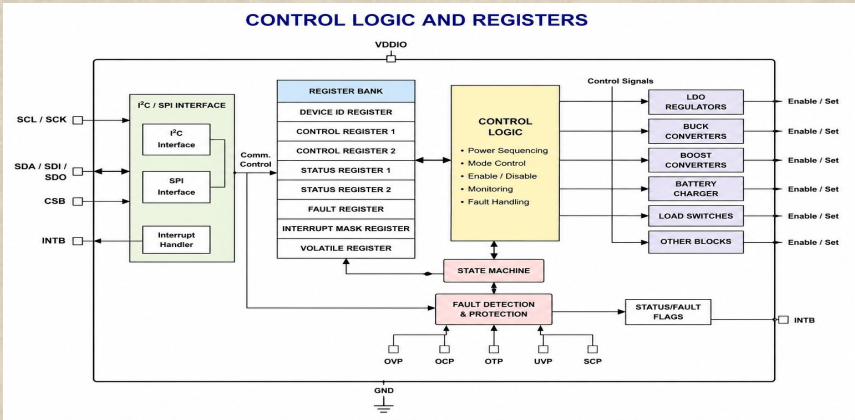


Fig 6. Control logic & Registers Block Diagram; [Ref: Chatgpt]

e) Low Dropout Voltage Regulator

A **Low Dropout (LDO) Regulator** is a linear voltage regulator that provides a stable output voltage even when the input voltage is only slightly higher than the output voltage. It is widely used in PMICs to supply low-noise power to sensitive analog and digital circuits.

Design

I. Reference Voltage

The bandgap reference generates a constant reference voltage: $V_{REF} = 1.2\ V$

II. Feedback Voltage

The output voltage is divided by the resistor network **R1** and **R2**:

$$V_{FB} = V_{OUT} \times \frac{R_2}{R_1 + R_2}$$

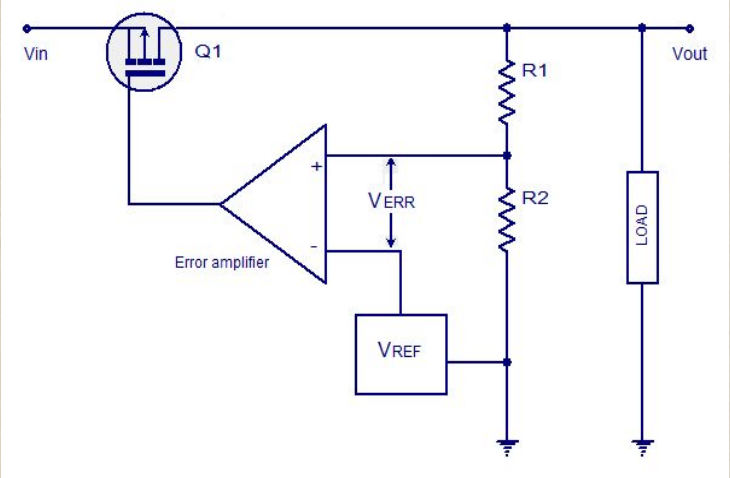


Fig 7. LDO Regulator Circuit; *Ref : Circuits today*

III. Error Voltage

The error amplifier compares the feedback voltage with the reference voltage:

$$V_{ERR} = V_{REF} - V_{FB}$$

IV. Pass Transistor Control

- If $V_{FB} < V_{REF}$, the error amplifier increases the gate drive of **Q1**, increasing the current to the load and raising V_{OUT}.
- If $V_{FB} > V_{REF}$, the amplifier reduces the gate drive, decreasing the current and lowering V_{OUT}.

V. Regulated Output Voltage

At steady state, $V_{FB} = V_{REF}$

$$V_{OUT} = V_{REF}\left(1 + \frac{R_1}{R_2}\right)$$

V. Dropout Voltage

The minimum input voltage required for regulation is :

$$V_{IN(MIN)} = V_{OUT} + V_{DO}$$

where V_{DO} is the **dropout voltage** (typically 50–300 mV for modern LDOs).

Switching Regulators

f) Buck Converter

A **Buck Converter** is a switching regulator that converts a higher DC input voltage into a lower regulated DC output voltage with high efficiency.

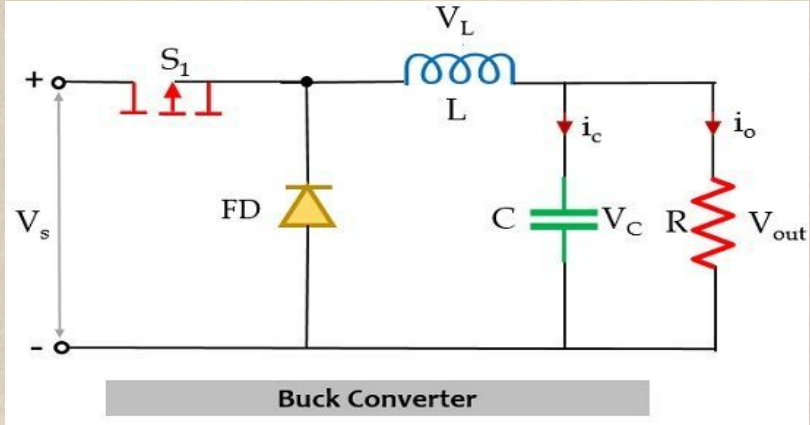


Fig 8. Buck Converter; *Ref: Electronics coach*

Design

I. Switch ON (S1 Closed)

- The MOSFET **S1** turns ON.
- Current flows from **Vs → S1 → L → Load (R)**.
- The **inductor (L)** stores energy.
- The **capacitor (C)** charges and smooths the output voltage.
- The **diode (FD)** is reverse biased.

$$V_L = V_s - V_{out}$$

Inductor voltage

II. Switch OFF (S1 Open)

- The MOSFET **S1** turns **OFF**.
- The inductor releases its stored energy.
- The freewheeling diode (FD) becomes forward biased.
- Current continues flowing through **L → C → R**, maintaining the output voltage.
- The capacitor supplies the load during switching.

$$V_L = -V_{out}$$

Inductor voltage

III. Output Voltage

For an ideal buck converter, ***Vout = D x Vs*** *; D = Duty Cycle, Vs = Input Voltage*

$$D = \frac{V_{out}}{V_{in}}$$

$$R = \frac{V_{out}}{I_{out}}$$

R = Load resistance (Ω)
Vout = Output voltage (V)
Iout = Output current (A)

$$C = \frac{\Delta I_L}{8 f_s \Delta V_{out}}$$

L = Inductor
ΔIL = Inductor ripple current
Fs = Switching frequency

$$L = \frac{(V_{in} - V_{out})D}{\Delta I_L f_s}$$

C = Output capacitor
ΔVout = Output voltage ripple
D= Duty Cycle

g) Boost Converter

A **Boost Converter** is a switching regulator that converts a low DC input voltage into a higher regulated DC output voltage with high efficiency.

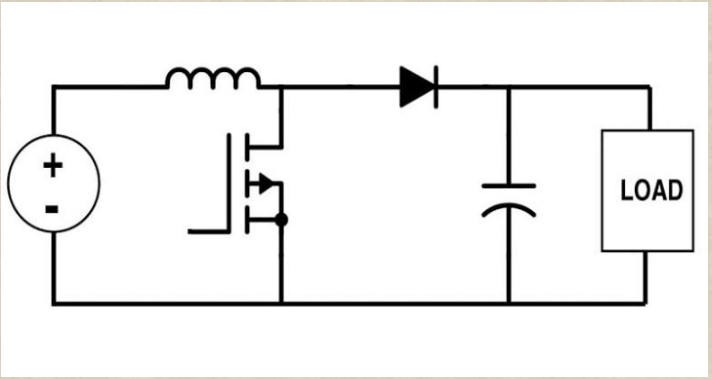


Fig 9. Boost Converter; [Ref : All about circuits]

Design

I. Switch ON (MOSFET Closed)

- The MOSFET turns **ON**.
- Current flows from **Vin** through the **inductor (L)** and MOSFET to ground.
- The **inductor stores energy** in its magnetic field.
- The **diode is reverse biased**, so the capacitor supplies the load.

$$V_L = V_{in}$$

Inductor voltage

II. Switch ON (MOSFET Closed)

- The MOSFET turns **OFF**.
- The inductor releases its stored energy.
- The **diode becomes forward biased**.
- The inductor voltage adds to the input voltage and charges the capacitor.
- The load receives power from both the inductor and capacitor.

$$V_L = V_{in} - V_{out}$$

Inductor voltage

$$D = 1 - \frac{V_{in}}{V_{out}}$$

D = Duty Cycle

$$L = \frac{V_{in} \times D}{\Delta I_L \times f_s}$$

L = Inductor
ΔIL = Inductor ripple current
Fs = Switching frequency

$$C = \frac{I_{out} \times D}{f_s \times \Delta V_{out}}$$

C = Output capacitor
ΔVout = Output voltage ripple
Iout = Output Current

h) Power Sequencer

- A **Power Sequencer** is a PMIC block that controls the **order and timing** of turning power rails ON and OFF.
- It ensures that different system components receive power in the correct sequence, preventing malfunction or damage.

i) Protection & Monitoring

- Continuously monitors voltage, current, and temperature.
- Detects overvoltage, undervoltage, overcurrent, overheating, and short circuits.
- Protects the PMIC and ensures safe, reliable operation.

j) Output Power Rails

Output Power Rails are regulated voltage outputs that supply power to the processor, memory, sensors, display, RF modules, and peripherals, providing the required voltage and current for each load.

08 Overall Working of PMIC

1. **Input Protection:** Protects against voltage, current, and ESD faults.
2. **Power Selector:** Selects battery or external power source.
3. **Battery Charger & Management:** Charges and monitors the battery safely.
4. **Bandgap Reference:** Provides a stable reference voltage.
5. **Control Logic & Registers:** Controls PMIC operation and communicates with the processor.
6. **Buck, Boost & LDO Regulators:** Step down or step up the input voltage as needed.
7. **Power Sequencer:** Enables power rails in the correct sequence.
8. **Protection & Monitoring:** Detects faults and ensures safe operation.
9. **Output Power Rails:** Supply regulated power to the CPU, memory, sensors, RF, and other components

09 Applications

- Used in **smartphones** for safe battery charging and power management.
- Used in **laptops** to improve battery life and charging efficiency.
- Used in **tablets** to monitor battery health and prevent overcharging.
- Used in **wearable devices** for efficient low-power battery operation.
- Used in **IoT devices** to extend battery life and reduce power consumption.
- Used in **electric vehicles (EVs)** to monitor and protect battery packs.

10 Design Challenges

- Achieving **high power efficiency** while minimizing power loss.
- Reducing **power consumption** to extend battery life.
- Maintaining **stable output voltage** under varying load conditions.
- Managing **heat dissipation** and preventing overheating.
- Minimizing **output voltage ripple and electrical noise**.

11 Future Scope

- Development of ultra-low-power PMICs for IoT and wearable devices.
- Integration of AI-based power management techniques.
- Faster charging and higher-efficiency battery management.
- Advanced PMICs for electric vehicles (EVs) and renewable energy systems.
- Adoption of smaller semiconductor process technologies for compact designs.

12 Conclusion

- PMICs integrate multiple power management functions into a single chip.
- They provide efficient voltage regulation, battery management, and system protection.
- PMICs improve power efficiency, battery life, and overall system reliability.
- They reduce power consumption, chip size, and system complexity.
- PMICs are widely used in smartphones, IoT devices, laptops, and electric vehicles..

References

[1] *J. Ruan et al., "A Hybrid Buck-or-Boost Converter for Fast-Transient and Wide-Voltage-Range Applications With Continuous Output Delivery Current," IEEE J. Solid-State Circuits, vol. 60, no. 9, pp. 3329–3341, Sept. 2025.*

[2] *M. Shim et al., "Power Management Circuit Techniques for Miniature Biomedical Sensing Systems," IEEE Open J. Solid-State Circuits Soc., vol. 6, pp. 25–46, 2026.*

[3] *S. Mu and M. Wagih, "Integrated RF Energy Harvesting and Power Conversion Circuits: A Review and Roadmap for Wireless-Powered SoCs," IEEE J. Microwaves, vol. 5, no. 6, pp. 1191–1211, Nov. 2025.*

[4] *Y. Xu et al., "Digital Low-Dropout Regulator-Assisted Buck DC-DC Converter Achieving 68-mV Droop Voltage and 95.5% Efficiency," IEEE Solid-State Circuits Lett., vol. 8, pp. 193–196, 2025.*

[5] *J. Kim et al., "A Command-Aware Hybrid LDO With Ultra-Small Voltage Droop and Fast Settling Time for Power-Supply-Induced Jitter Mitigation in HBM Interfaces," IEEE J. Solid-State Circuits, vol. 61, no. 1, pp. 116–128, Jan. 2026.*

[6] *Y.-J. Lee et al., "A Fully Synthesizable, Fine-Grained Distributed Digital-LDO Fabric Featuring Freely Linkable Current-Sharing Network for Scalable On-Chip Power Delivery," IEEE J. Solid-State Circuits, vol. 61, no. 7, pp. 3688–3703, Jul. 2026.*



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